Question Bank – Artificial Intelligence

1. Explain the concept of Artificial Intelligence and its key features.
2. Describe different types of environments in AI.
3. Explain agent and environment with suitable examples.
4. Differentiate between rational agents and intelligent agents.
5. Describe the structure of an intelligent agent with a neat diagram.
6. Elaborate the structure of intelligent agents.
7. Describe types of intelligent agents.
8. Explain different types of agents in AI with architecture.
9. Explain the structure of problem-solving agents.
10. Elaborate problem-solving agents and their working mechanism.
11. Define problem formulation in AI with a suitable example.
12. Discuss problem formulation in Artificial Intelligence.
13. Illustrate the process of problem formulation in Artificial Intelligence using a suitable example.
14. Explain AI search techniques with examples.
15. Discuss AI techniques and search knowledge in detail.
16. Summarize the role of search knowledge in Artificial Intelligence problem solving.
17. Compare uninformed search and heuristic search techniques with suitable comparison parameters.
18. Explain the role of heuristic functions in search.
19. Explain heuristic search techniques.
20. Explain Breadth First Search with example.
21. Demonstrate the working of Breadth First Search using state-space traversal.
22. Differentiate between Breadth First Search and Depth First Search.
23. Explain Depth First Search with advantages and disadvantages.
24. Describe Hill Climbing algorithm with limitations.
25. Apply Hill Climbing algorithm to solve a search problem and identify its practical limitations.
26. Compare DFS and Hill Climbing algorithms.
27. Elaborate game playing in AI with suitable example.
28. Explain the problems faced in game playing and their solutions.
29. Assess major computational challenges encountered in adversarial game playing.
30. Define the minimax algorithm with diagram.
31. Apply the minimax framework to determine optimal strategic decisions in game playing.
32. Describe evaluation functions in adversarial search.
33. Illustrate the evaluation functions in game trees.
34. Examine the significance of evaluation functions in game tree search.
35. Discuss Alpha-Beta pruning with example.
36. Explain problem reduction in AO* algorithm.
37. Explain Bayesian Network with suitable example.
38. Explain Hidden Markov Models with applications.
39. Apply Hidden Markov Models for sequential reasoning with suitable illustration.
40. Analyze predictive estimation using Kalman filtering in dynamic systems.
41. Illustrate Dynamic Bayesian Networks for temporal probabilistic reasoning.
42. Analyze the architecture of neural networks and illustrate their role in computational problem solving.
43. Illustrate the concept of machine learning in AI.
44. Apply the principles of genetic algorithms to solve optimization problems with suitable process flow.
45. Examine the practical applications of fuzzy reasoning systems in handling uncertain real-world situations.
46. Illustrate the working of perception mechanisms in cognitive robotics with suitable examples.
47. Demonstrate perception mechanisms used in cognitive robotics with suitable examples.
48. Analyze localization and mapping techniques used for autonomous robotic navigation.
49. Evaluate ethical risks associated with intelligent robotic systems in real-world deployment.
50. Illustrate ethics and risks of Artificial Intelligence.
51. Explain AND-OR graphs with a suitable example.

Question Bank – Artificial Intelligence

52. What is a search tree? How is it different from a state space graph?
53. What is the difference between a Markov chain and a Hidden Markov Model?
54. Show that Kalman Filter is a special case of a Dynamic Bayesian Network.
55. Define Bayesian Network. What does each node and edge represent in it?
56. Define a heuristic function. Give one example of a heuristic used in path finding.
57. Define a rational agent. What is the difference between a rational agent and a random agent?
58. What is an agent in AI? Give two real-life examples of an agent.
59. What is a game tree? Draw a simple game tree with two levels.
60. What is machine learning? Name the three types of machine learning.

FOCUSED TOPICS:

1. Agents & Environment

- *Types of agents with architecture*
- *Rational vs intelligent agents*
- *Types of environments*

2. Problem Solving

- *Structure of problem-solving agents*
- *Problem formulation with example*

3. Search Algorithms

- *BFS with example*
- *DFS with advantages & disadvantages*
- *Hill Climbing with limitations*
- *Uninformed vs Heuristic search*

4. Adversarial Search

- *Minimax algorithm with diagram*
- *Alpha-Beta pruning with example*
- *Evaluation functions in game trees*
- *Game playing problems & solutions*

5. Probabilistic Reasoning

- *Bayesian Network with example*
- *Hidden Markov Models with applications*
- *Kalman Filter basics*
- *Dynamic Bayesian Networks*

6. Advanced Topics

- *Neural Network architecture*
- *Genetic Algorithms*
- *Fuzzy Reasoning Systems*

7. Robotics & Ethics

- *Perception in cognitive robotics*
- *Localization & mapping (SLAM)*
- *Ethics & risks of AI*